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ABCPMX SOFTWARE V1.00 COMPOSITE BEAMS DESIGN

VALIDATION REPORT

« Etablissement certifié qualité ISO 9001, le CTICM assure un suivi de
chaque étude dans le plus strict respect de ses procédures qualité »

Date :	07/10/2024	Auteur :	GUD+POM	Nombre total de pages : (y compris celle-ci)	70
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	Research and Valorisation Department		
	Project: ABCPMX-II Software - DAL-22-003		
	Product reference: DRV/PMX/MV/001	Rév. :	B

Révision	Date	Auteur	Objet
A	07/10/2024	GUD+POM	First release for V1.00
B	09/01/2025	POM	Introduction of new tests C06 and C07
C			
D			
E			

Date :	07/10/2024	Auteur :	GUD+POM	Nombre total de pages : (y compris celle-ci)	70
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1 OBJECT

ABC-PMX software aims at the calculation of composite beams according to EN 1994-1-1. Its development is co-funded by ArcelorMittal and CTICM. This document constitutes the Validation Manual.

2 REFERENCES

2.1 First generation of Eurocodes

- [1] EN 1990 – Basic of structural design. AFNOR. March 2003.
- [2] EN 1991-1-2 - Eurocode 1: Actions on structures Part 1-2: General actions — Actions on structures exposed to fire
- [3] EN 1992-1-1 – Design of concrete structures. Part 1-1: General rules and rules for buildings. AFNOR. October 2005.
- [4] EN 1993-1-1 – Design of steel structures. Part 1-1: General rules and rules for buildings. AFNOR. October 2005.
- [5] EN 1993-1-2 – Design of steel structures. Part 1-2: General rules - Structural fire design
- [6] EN 1993-1-5 – Design of steel structures. Part 1-5: Plated structural elements. AFNOR. March 2007.
- [7] EN 1994-1-1 – Design of composite steel and concrete structures. Part 1-1: General rules and rules for buildings. AFNOR. June 2005.
- [8] EN 1994-1-2 – Design of composite steel and concrete structures. Part 1-2: General rules - Structural fire design

2.2 Drafts for the second generation of Eurocodes

- [9] prEN 1992-1-1 – Design of concrete structures. Part 1-1: General rules and rules for buildings. Document SC2N2034
- [10] FprEN 1993-1-1 – Design of steel structures. Part 1-1: General rules and rules for buildings. Document SC3 N3504 – 26/11/21
- [11] FprEN 1993-1-5 – Design of steel structures. Part 1-5: Plated structural elements. Document SC3 N3728 – 26/11/21
- [12] prEN 1994-1-1 – Design of composite steel and concrete structures. Part 1-1: General rules and rules for buildings. CEN/TC250/SC4 document N 2394. November 2021 - Draft.

2.3 Scientific references

- [13] Design of floor structures for human induced vibrations. M. Feldmann, Ch. Heinemeyer, Ch. Butz, E. Caetano, A. Cunha, F. Galanti, A. Goldack, O. Hechler, S. Hicks, A. Keil, M. Lukic, R. Obiala, M. Schlaich, G. Sedlacek, A. Smith, P. Waarts – Background document in support to the implementation, harmonization and further development in the Eurocodes – Joint Report – 2009 – JRC Scientific and Technical Reports
- [14] European Recommendations for the Application of Metal Sheeting acting as a Diaphragm – Stresses Skin Design – ECCS European Convention for Constructional Steelwork – Report no 88 – 1995
- [15] MARTIN P-O., RODIER A. - Design of the web-to-flange welds of a steel and concrete composite beam with a welded steel section – Revue Construction Métallique 1/2023
- [16] MARTIN P-O., NGUYEN T-M. - Resistance to lateral torsional buckling of a secondary beam in construction phase – Restraint by the steel deck – Revue Construction Métallique 3/2022
- [17] Déversement des poutres en acier – Collection CTICM (in French)
- [18] Guide de conception pour les planchers minces à poutres incorporées - ECCS European Convention for Constructional Steelwork – Report no 83 – 1996
- [19] COSFB Software – Technical specifications – 15/09/2020 – ArcelorMittal

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[20] Nomogrammes – Méthode graphique pour déterminer la résistance au feu des structures en acier, selon l'EN 1993-1-2 : 2005 – Centre Information Acier

3 PRINCIPLES OF VALIDATION

This report includes test cases for all configurations. For each of them, the following are specified:

- the input data of the test case ;
- software output data ;
- the results obtained by the reference ;
- the comparison between the software output results and those obtained by the reference.

The input data is strictly identical between the software and the reference.

The references values used for software validation come from Excel spreadsheets developed specifically at CTICM and scientific literature references.

The comparison of the results is based on the evaluation of the differences between the values provided by the software and those of the reference. This comparison will be limited to the main results (verification ratios). The intermediate results will therefore be neither detailed nor compared here.

The analysis of the results obtained and the deviations observed make it possible to confirm that the ABC-PMX-II software produces values in accordance with the calculation methods applied.

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4 SCOPE GRID

4.1 Normal situations

Parameters	Test C01	Test C01b	Test C02	Test C03	Test C04
Type of section / beam	Composite beam	Composite beam	Composite beam	Composite beam / partial encasement	Composite beam / partial encasement
Type of profile	HR	HR	HR	HR	HR
Steel grade	S275 (EC3)	S275 (EC3)	S275 (EC3)	S355 M/ML	S355 EC3
Construction phase	No	yes	Yes	Yes	No
Propping	unpropped	unpropped	unpropped	unpropped	unpropped
Cantilever spans	No	No	No	No	No
Type of slab	Composite	Composite	Composite	Composite	Composite
Steel deck label	Cofraplus 60	Cofraplus 60	Cofraplus 60	Cofraplus 60	Cofraplus 60
Ribs orientations	Perpendicular	Perpendicular	Perpendicular	Perpendicular	Perpendicular
Restraint by the deck	-	-	Yes	No	No
Studs	19-100	19-100	-	-	19-100
Standard	Generation 1	Generation 1	Generation 1	Generation 1	Generation 1
Type of design	Plastic	Plastic			

Parameters	Test C05	Test C05b	Test C06	Test C07	
Type of section / beam	Composite beam	Composite beam	Composite beam	Composite beam	
Type of profile	HR	HR	HR	HR	
Steel grade	S355 JR	S355 JR	S460	S420	
Construction phase	No	No	No	No	
Propping	unpropped	unpropped	propped	propped	
Cantilever spans	left	right	no	no	
Type of slab	Composite	Composite	Solid	Composite	
Steel deck label	Cofraplus 60	Cofraplus 60	/	Cofraplus 220	
Ribs orientations	Perpendicular	Perpendicular	/	Perpendicular	
Restraint by the deck	-	-	/	-	
Studs	19-100	19-100	22-125	22-100	
Standard	Generation 1	Generation 1	Generation 1	Generation 1	
Type of design	Plastic	Plastic	Plastic	Plastic	

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Parameter	Test S01	Test S02	Tests S03, S03b	Tests S04, S04b	
Type of section / beam	Steel beam	Steel beam	Steel beam	Steel beam	
Type of profile	HR	Welded	Welded unsymmetrical	Welded unsymmetrical	
Steel grade	S355 M/ML	S355 JR	S275 JR	S460 EC3	
Construction phase	-	-	-	-	
Propping	-	-	-	-	
Cantilever spans	Left and right	No	Left or right	Left	
Type of slab	Solid	Solid	Solid	Solid	
Steel deck label	-	-	-	-	
Ribs orientations	-	-	-	-	
Restraint by the deck	-	-	-	-	
Studs	-	-	-	-	
Standard	Generation 1	Generation 1	Generation 1	Generation 1	
Type of design	Plastic	Plastic	Plastic	Elastic	
Shear buckling	No	Yes	No	Yes	
Interaction MV	No	No	Yes	Yes	
Interaction M shear buckling	No	No	No	Yes	

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Parameters	Test F05b	Test F06
Type of section / beam	Composite beam	Composite beam partially encased
Type of profile	Welded	HR
Fire protection	Spray	Partial encasement
Steel grade	S355	S355
Cantilever spans	No	No
Type of slab	Solid	Composite
Steel deck label		Cofraplus 60
Ribs orientations		Perpendicular
Studs	22x125	19x100
Standard	Generation 1	Generation 1
Heating of the slab	Numerical	Standard

5 TESTS CASES – COMPOSITE BEAMS

5.1 CASE C01: Single span composite beam

5.1.1 Description

This test is based on the *Construction Métallique* review n°2018-2. It deals with a single span composite beam with a composite slab.

The reference values are taken from the paper. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

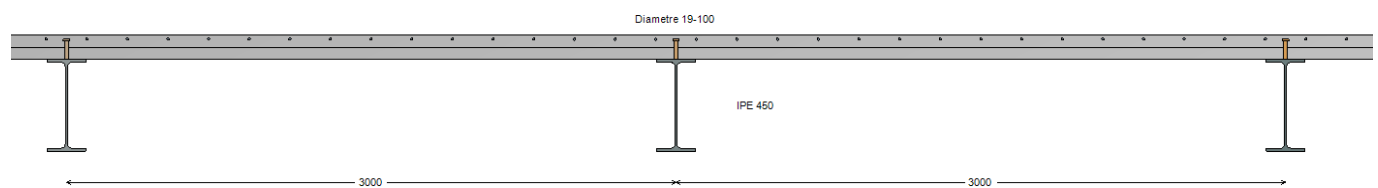
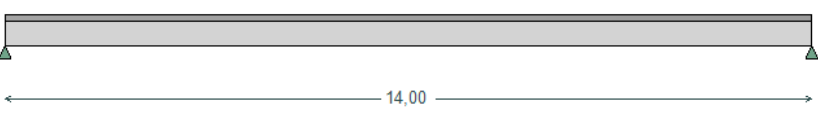
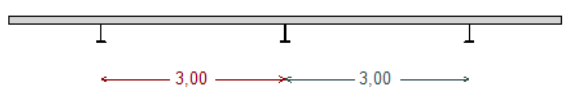


Figure 1 : Transversal view

5.1.2 Parameters

Definition of the beam:

Span lengths Main span L = 14 m ☐ Left cantilever span ☐ The slab is continuous at the left support ☐ Right cantilever span ☐ The slab is continuous at the right support		
Spacings ☒ Intermediate beam ☐ Edge beam Spacings $d_1 = 3$ m $d_2 = 3$ m	Slab openings ☐ Left opening ☐ Right opening	

Definition of the slab:

Composite slab

Total depth: $t_d = 120$ mm

Perpendicular ribs

Steel profiled sheeting: Cofraplus 60, thickness 0,75 mm

Prepunched sheeting

Definition of the profile:

Hot rolled profile

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Profile: IPE 450
Steel grade: S275 EC3

Definition of the concrete:

Normal concrete
Concrete class: C25/30
Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 19x100
One stud in each rib
Unpropped beam during construction

Definition of loads:

G1: self-weight only
G2: surface load $1,4 \text{ kN/m}^2$
live load Q1: surface load $2,5 \text{ kN/m}^2$

Combination of loads:

ULS: $1,35 (G1+G2) + 1,5 Q1$
SLS: $(G1+G2) + Q1$

Design parameters:

Standard: Eurocodes first generation
Design: According to the class of cross-section
Relative humidity: 50 %
Shrinkage deformation: 325×10^{-6}
Age of concrete: $t_0 = 28 d$ for G2
 $t = 50 \times 365 d$

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5.1.3 Verification of results

CASE C01 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	98.8	98.8	
Shear area : $A_v =$ (cm ²)	50.9	50.8	0.20%
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	33 740	33 743	0.01%
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 676	1 672	0.24%
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 500	1 500	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	176	176	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 702	1 702	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	276	272.8	1.17%
Torsional constant : $I_t =$ (cm ⁴)	66.66	66.75	0.13%
Warping constant : $I_w =$ (cm ⁶)	780 739	791 005	1.30%
Mass by length unit : $m_{lin} =$ (kg/ml)	77.6	77.6	
Total mass : $m =$ (kg)	1086	1 086	
Total paint surface : $S =$ (cm ²)	224 700	224 721	0.01%
Paint surface (without upper face) : $S =$ (cm ²)	198 100	198 121	0.01%
Massiveness : $M =$ (m ⁻¹)	162	162.4	0.25%
Massiveness (without upper face) : $M =$ (m ⁻¹)	143	143.2	0.14%
Nominal yield strenght : $f_y =$ (MPa)	275	275	
Young's modulus : $E =$ (MPa)	210 000	210 000	
Density : $\rho =$ (kg/m ³)	7 850	7 850	
Tensile resistance : $N_{t,Rd} =$ (kN)	2 717	2 718	0.04%
Shear resistance : $V_{pl,Rd} =$ (kN)	807	807	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	468	468	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	412.5	412	0.12%
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	44,8	44,8	
Shear buckling resistance : $72\varepsilon/\eta =$	66,5	66,5	
Verification of web buckling required?	No	No	

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CASE C01 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	124	124	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	44.37	44.37	
Shear resistance of the connector : $P_{Rd} =$ (kN)	52.9	52.9	
Reduction factor : $k_t =$	0.71	0.71	

CASE C01 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : $b_{eff} =$ (m)	3,0	3,0	
Modular ratio : $n(G_2) =$	27.5	27.5	
Modular ratio : $n(Q_1) =$	6.67	6.67	
Modular ratio : $n(SHC) =$	26.1	26.1	
Second moment of area (y-y axis) : $I_{el,y}(n = 6.67)$ (cm ⁴)	106 570	106 571	
Second moment of area (y-y axis) : $I_{el,y}(n = 26.1)$ (cm ⁴)	74 820	74 821	
Second moment of area (y-y axis) : $I_{el,y}(n = 27.5)$ (cm ⁴)	73 533	73 534	
Second moment of area (y-y axis) : $I_{el,y}(steel)$ (cm ⁴)	33 742	33 743	
Elastic neutral axis : $z_{el,y}(n = 6.67)$ (mm)	6.83	6.82	0.15%
Elastic neutral axis : $z_{el,y}(n = 26.1)$ (mm)	-93.4	-93.4	
Elastic neutral axis : $z_{el,y}(n = 27.5)$ (mm)	-97.5	-97.5	
Elastic neutral axis : $z_{el,y}(steel)$ (mm)	-225	-225	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	846	846	

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CASE C01 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	50.8	50.8	
Bending moment : $M(G_1) =$ (kN.m)	178	178	
Shear force : $V(G_2) =$ (kN)	29.4	29.4	
Bending moment : $M(G_2) =$ (kN.m)	103	103	
Shear force : $V(Q_1) =$ (kN)	52.5	52.5	
Bending moment : $M(Q_1) =$ (kN.m)	184	184	
Shear force : $V(Q_c) =$ (kN)	15	15	
Bending moment : $M(Q_c) =$ (kN.m)	64.9	64.9	

CASE C01 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.84	0.84	
Resistance to shear force : $\Gamma_V =$	0.23	0.23	
Resistance to MV interaction : $\Gamma_{MV} =$	0.84	0.84	
Degree of shear connection : $\eta =$	0.68	0.68	
Minimum degree of connection : $\eta_{\min} =$	0.57	0.57	
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2.06	2.06	
Min. compression strut inclination angle : $\theta_{f,\min} =$ (°)	27	27	
Resistance of the compression strut : $\Gamma_{sf} =$	0.57	0.57	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	0	0	

CASE C01 RESULTS: ULS AT CONSTRUCTION STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.72	0.72	
Resistance to shear force : $\Gamma_V =$	0.11	0.11	

CASE C01 RESULTS: SLS ASSUMING FULL SHEAR CONNECTION	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	51.2	51.2	
Deflection : $\delta(G_2)$ (mm)	13.6	13.6	
Deflection : $\delta(Q_1)$ (mm)	16.8	16.8	
Deflection : $\delta(SHC)$ (mm)	13.8	13.9	0.72%
Deflection : $\delta(SLS)$ (mm)	95.5	95.5	
Eigen frequency : $f(G + 0.2Q_1)$ (Hz)	3.3	3.3	

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5.2 CASE C01b: Simply supported composite beam

5.2.1 Description

This test is based on the *Construction Métallique* review n°2018-2. It deals with a single span composite beam with a composite slab in the construction phase.

It corresponds to the construction phase of the Case C01.

5.2.2 Parameters

All parameters are the same as in Case C01, with the additional parameters:

Lateral restraint of the profile:

No lateral restraints in the span

No restraint by the steel profiled sheeting

Definition of loads:

G1: self-weight only

live load QC: surface load 0,5 kN/m²

distributed load : 3 kN/m on the 3 m at the center of the span

Combination of loads:

ULS: 1,35 G1 + 1,5 QC

5.2.3 Verification of results

CASE C01b RESULTS: ULS AT CONSTRUCTION STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.72	0.72	
Resistance to shear force : $\Gamma_V =$	0.11	0.11	

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5.3 CASE C02: Single span composite beam laterally restrained during its construction phase

5.3.1 Description

This test is based on the *Construction Métallique* review n°2022-2. It deals with a single span composite beam which is laterally restrained by the steel deck during its construction phase.

The reference values are taken from the paper. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

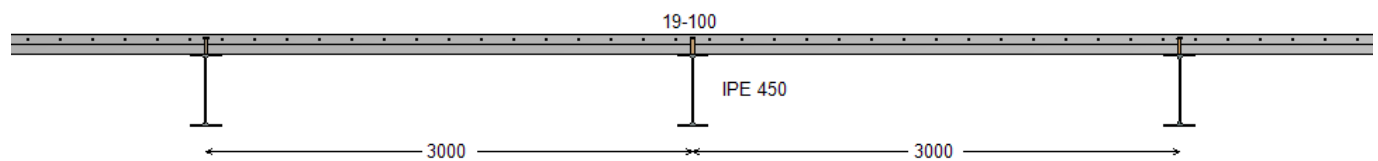


Figure 2 : Transversal view

5.3.2 Parameters

All parameters are the same as in Case C01, with the additional parameters:

Lateral restraint of the profile:

No lateral restraints in the span

Restraint by the steel profiled sheeting:

Paramètres du sol	Fixation du bac à la poutre	Paramètres
☒ Poutre est maintenue latéralement par le bac Nombre de tôles dans la direction transversale: 2 Dimensions du sol (travée principale) Longueur de travée: 14,00 m Largeur: 12,00 m Type de transition entre deux tôles adjacentes: Tôles adjacentes sans jeu	Une attache: à chaque onde Type d'attaches: Fixations par pistolet à cartouches Diamètre: De 3,7 à 4,8 mm Glissement: 0,10 mm/kN	Rigidité au cisaillement K = 0,176 α_5 = 0,500 c ₁₁ = 0,824 mm/kN c ₁₂ = 0,120 mm/kN c ₂₁ = 0,064 mm/kN c ₂₂ = 0,064 mm/kN c = 1,073 mm i _{act} = 13049 kN/m
Tôle individuelle Nombre de travée traversée par les tôles: 2 Dimensions Longueur: 6,00 m Largeur: 1,04 m	Couturage Type d'attaches: Vis autoforeuses Diamètre: De 4,1 à 4,8 mm Glissement: 0,25 mm/kN Espacement: 500 mm	Rigidité à la flexion k _{theta,A} = 18,2 kN.m/m k _{theta,C} = 186 kN.m/m k _{theta} = 16,5 kN.m/m ☒ Prise en compte de la rigidité flexionnelle du bac

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Definition of loads:

G1: self-weight only
live load QC: surface load 0,5 kN/m²
distributed load : 3 kN/m on the 3 m at the center of the span

Combination of loads:

ULS: 1,35 G1 + 1,5 QC

5.3.3 Verification of results

CASE C02 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	50.8	50.8	
Bending moment : $M(G_1) =$ (kN.m)	178	178	
Shear force : $V(Q_c) =$ (kN)	15	15	
Bending moment : $M(Q_c) =$ (kN.m)	64.8	64.9	
Shear force : $V_{Ed} =$ (kN)	91	91	
Bending moment : $M_{Ed} =$ (kN.m)	337	337	

CASE C02 RESULTS: ULS during construction stage	Reference	ABCPMX	Dev.
Critical moment to lateral torsional buckling (without flexural bending of deck accounted for) : M_{cr} (kN.m)	2 260	2 304	-1.91%
Critical moment to lateral torsional buckling (with flexural bending of deck accounted for) : M_{cr} (kN.m)	3 237	3 299	-1.88%
Resistance to lateral torsional buckling (with flexural bending of deck accounted for) : $\Gamma_{LT} =$	0.77	0.77	

5.4 CASE C03: Single span composite beam with partial encasement (construction phase)

5.4.1 Description

This test is based on the paper in the *Construction Métallique* review n°2023-3. It deals with a single span composite beam with a composite slab and partial encasement during its construction phase.

The values from the paper are used as references. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

It corresponds to the construction phase of the case C04.

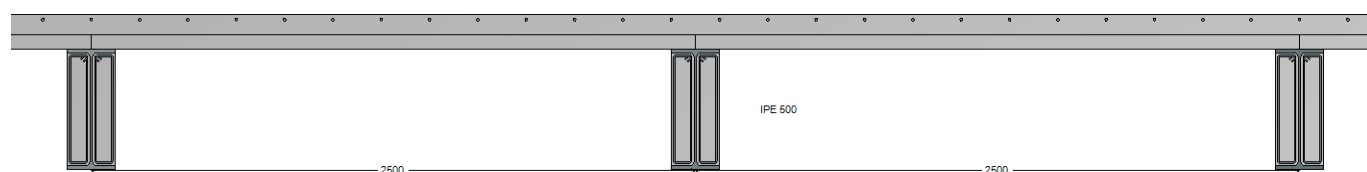
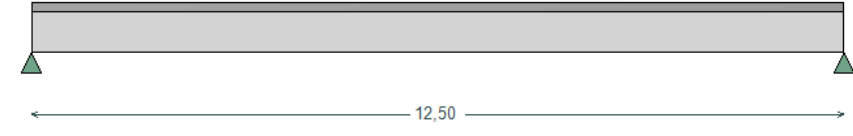
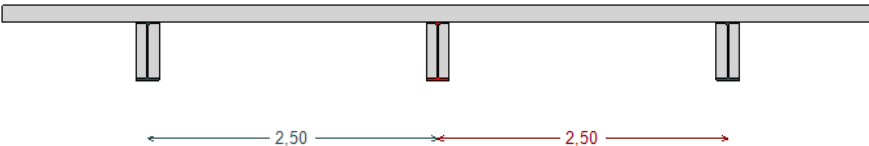


Figure 3 : Transversal view

5.4.2 Parameters

Definition of the beam:

Span lengths Main span $L = 12,50$ m ☐ Left cantilever span ☐ Right cantilever span	
Spacings ☒ Intermediate beam ☐ Edge beam Spacings $d_1 = 2,5$ m $d_2 = 2,5$ m	

Definition of the slab:

Composite slab

Total depth: $t_d = 140$ mm

Perpendicular ribs

Continuous ribs with un-continuous deck

Steel profiled sheeting: Cofraplus 60, thickness 1 mm

Prepunched sheeting

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Unpropped beam during construction

Definition of the profile:

Hot rolled profile
Profile: IPE A 500
Steel grade: S355 M/ML

Definition of the concrete:

Normal concrete
Concrete class: C25/30
Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the partial encasement:

Dimensions

Width

$b_c = 1$

x b_y

$b_c = 200$ mm

Stirrups

Type

Closed stirrups

Diameter

$\phi_e = 6$ mm

Coverage

$u_y = 10$ mm

$u_z = 48$ mm

Concrete

Class

C25/30

$f_{ck} = 25$ MPa

$E_{cm} = 31,476$ GPa

Steel of the reinforcement

Class

B500

$f_{sk} = 500$ MPa

Longitudinal reinforcement

External

Middle

Internal

Number

Single

None

Single

Diameter

8

8

12

mm

Active rebars

☒

☐

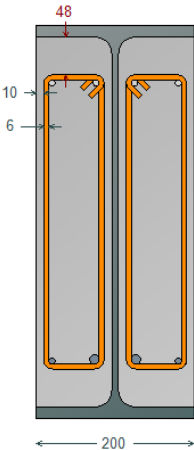
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Position

$z_{s,inf} = 423$ mm

Lower layer

$A_{s,inf} = 163,363$ mm²



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Lateral restraint of the profile:

No lateral restraints in the span

No restraint by the steel profiled sheeting

Definition of loads:

G1: self-weight only

live load QC: distributed load: 1,86 kN/m on the full length

distributed load: 1,05 kN/m on the 3 m at the centre of the span

Combination of loads:

ULS: 1,35 G1 + 1,5 QC

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5.4.3 Verification of results

CASE C03 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	101	101.1	
Shear area : $A_v =$ (cm ²)	50.4	50.4	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	42 930	42 933	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 949	1 936	0.18%
Elastic section modulus about y-y axis : $W_{el,y} =$ (cm ³)	1 728	1 728	
Elastic section modulus about z-z axis : $W_{el,z} =$ (cm ³)	193,9	193.6	0.18%
Plastic section modulus about y-y axis : $W_{pl,y} =$ (cm ³)	1 946	1 946	
Plastic section modulus about z-z axis : $W_{pl,z} =$ (cm ³)	301.6	298.3	1.15%
Torsional moment : $I_t =$ (cm ⁴)	64.18	64.28	-0.16%
Warping moment : $I_w =$ (cm ⁶)	1 125 000	1 125 230	
Mass by length unit : $m_{lin} =$ (kg/ml)	79,4	79,4	
Mass : $m =$ (kg)	994	992	0.16%
Total paint surface : $S =$ (cm ²)	217 643	217 643	
Paint surface (without upper face) : $S =$ (cm ²)	192 643	192 643	
Section factor : $M =$ (m ⁻¹)	172.0	172.2	-0.14%
Section factor (without upper face) : $M =$ (m ⁻¹)	152.2	152.4	-0.13%
Nominal yield strength : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	3 595	3 589	0.15%
Shear resistance : $V_{pl,Rd} =$ (kN)	1 036	1 033	0.32%
Plastic resistance to bending : $M_{pl,Rd} =$ (kN.m)	692	691	0.16%
Elastic resistance to bending : $M_{el,Rd} =$ (kN.m)	614	613	0.24%
Section class for pure bending :	1	1	
Section class for pure compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	50.7	50.7	
Shear buckling resistance : $124\varepsilon =$	101	101	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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CASE C03 RESULTS: PARTIAL ENCASEMENT	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	191	191	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	

CASE C03 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	210	210	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	60.08	60.08	

CASE C03 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Modular ratio : $n_{enrobage}(G_I) =$	23.7	23.7	
Modular ratio : $n_{enrobage}(Q_C) =$	6.67	6.67	
Second moment of area (y-y axis) : $I_{el,y}(n = 6.67)$ (cm ⁴)	53 002	52 921	0.15%
Second moment of area (y-y axis) : $I_{el,y}(n = 23.7)$ (cm ⁴)	47 200	47 128	-0.15%
Second moment of area (y-y axis) : $I_{el,y}(n = 24.7)$ (cm ⁴)	47 083	47 005	-0.17%
Elastic neutral axis : $z_{el,y}(n = 6.67)$ (mm)	205	205	
Elastic neutral axis : $z_{el,y}(n = 23.7)$ (mm)	235	235	
Elastic neutral axis : $z_{el,y}(n = 24.7)$ (mm)	236	236	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	777	775	0.26%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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CASE C03 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	60.5	60.5	
Bending moment : $M(G_1) =$ (kN.m)	189	189	
Shear force : $V(Q_c) =$ (kN)	13.3	13.3	
Bending moment : $M(Q_c) =$ (kN.m)	45.3	45.4	-0.22%

CASE C03 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.42	0.42	
Resistance to shear force : $\Gamma_V =$	0.10	0.10	
Resistance to MV interaction : $\Gamma_{MV} =$	0.42	0.42	
Critical moment to lateral torsional buckling : M_{cr} (kN.m)	903	914	-1.20%
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	0.66	0.63	4.76%

CASE C03 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	31.0	31.1	-0.32%
Deflection : $\delta(Q_c)$ (mm)	6.5	6.5	
Deflection : $\delta(SLS)$ (mm)	37.5	37.6	-0.27%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	09/01/2025	Authors:	GUD+POM	Page:	22	Rév. : B

5.5 CASE C04: Single span composite beam with partial encasement (composite phase)

5.5.1 Description

This test is based on the paper in the *Construction Métallique* review n°2023-4. It deals with a single span composite beam with a composite slab and partial encasement during its final phase in composite stage.

The values from the paper are used as references. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

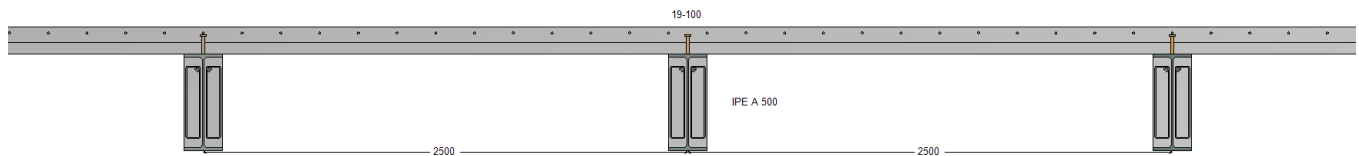


Figure 4 : Transversal view

5.5.2 Parameters

All parameters are the same as in Case 3, with the following specific parameters:

Definition of the connection:

Studs: 19x100
Two studs in each rib
Unpropped beam during construction

Definition of loads:

G1: self-weight only
G2: surface load 1,0 kN/m²
live load Q1: surface load 2,5 kN/m²

Combination of loads:

ULS: 1,35 (G1+G2) + 1,5 Q1
SLS: (G1+G2) + Q1

Design parameters:

Standard: Eurocodes first generation
Design: According to the class of cross-section
Relative humidity: 50 %
Shrinkage deformation: 325×10^{-6}
Age of concrete: $t_0 = 28 d$ for G2
 $t = 50 \times 365 d$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	09/01/2025	Authors:	GUD+POM	Page:	23	Rév. : B

5.5.3 Verification of results

CASE C04 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	101	101.1	
Shear area : $A_v =$ (cm ²)	50.4	50.4	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	42 930	42 933	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 949	1 936	0.18%
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 728	1 728	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	193,9	193.6	0.18%
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 946	1 946	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	301.6	298.3	1.15%
Torsional constant : $I_t =$ (cm ⁴)	64.18	64.28	-0.16%
Warping constant : $I_w =$ (cm ⁶)	1 125 000	1 125 230	
Mass by length unit : $m_{lin} =$ (kg/ml)	79,4	79,4	
Mass : $m =$ (kg)	994	992	0.16%
Total paint surface : $S =$ (cm ²)	217 643	217 643	
Paint surface (without upper face) : $S =$ (cm ²)	192 643	192 643	
Section factor : $M =$ (m ⁻¹)	172.0	172.2	-0.14%
Section factor (without upper face) : $M =$ (m ⁻¹)	152.2	152.4	-0.13%
Nominal yield strength : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	3 595	3 589	0.15%
Shear resistance : $V_{pl,Rd} =$ (kN)	1 036	1 033	0.32%
Plastic resistance to bending : $M_{pl,Rd} =$ (kN.m)	692	691	0.16%
Elastic resistance to bending : $M_{el,Rd} =$ (kN.m)	614	613	0.24%
Section class for pure bending :	1	1	
Section class for pure compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	50.7	50.7	
Shear buckling resistance : $124\varepsilon =$	101	101	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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CASE C04 RESULTS: PARTIAL ENCASEMENT	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	191	191	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	

CASE C04 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	164	164	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	60.08	60.08	
Shear resistance of studs : $P_{Rd} =$ (kN)	37.4	37.4	
Reduction factor : $k_t =$	0.50	0.50	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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CASE C04 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : b_{eff} = (m)	2.50	2.50	
Modular ratio : $n_{slab}(G_1)$ =	—	—	
Modular ratio : $n_{encasement}(G_1)$ =	23.7	23.7	
Modular ratio : $n_{slab}(G_2)$ =	26.6	26.6	
Modular ratio : $n_{encasement}(G_2)$ =	23.7	23.7	
Modular ratio : $n_{slab}(Q_1)$ =	6.67	6.67	
Modular ratio : $n_{encasement}(Q_1)$ =	6.67	6.67	
Modular ratio : $n_{slab}(Q_c)$ =	∞	∞	
Modular ratio : $n_{encasement}(Q_c)$ =	6.67	6.67	
Modular ratio : $n_{slab}(SHC)$ =	25.2	25.2	
Modular ratio : $n_{slab}(SHC)$ =	24.7	24.7	
Moment of inertia about y-y axis : $I_{el,y}(G_1)$ (cm ⁴)	47 237	47 128	0.23%
Moment of inertia about y-y axis : $I_{el,y}(G_2)$ (cm ⁴)	99 895	99 778	0.12%
Moment of inertia about y-y axis : $I_{el,y}(Q_1)$ (cm ⁴)	142 875	142 812	0.14%
Moment of inertia about y-y axis : $I_{el,y}(Q_c)$ (cm ⁴)	53 094	52 921	0.33%
Moment of inertia about y-y axis : $I_{el,y}(SHC)$ (cm ⁴)	101 577	101 471	0.10 %
Elastic neutral axis : $z_{el,y}(G_1)$ (mm)	-235	-235	0.00%
Elastic neutral axis : $z_{el,y}(G_2)$ (mm)	-102.4	-102	0.39%
Elastic neutral axis : $z_{el,y}(Q_1)$ (mm)	9.38	9.51	-1.37%
Elastic neutral axis : $z_{el,y}(Q_c)$ (mm)	-205	-205	0.00%
Elastic neutral axis : $z_{el,y}(SHC)$ (mm)	-98.1	-97.9	0.20%
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	1 190	1 243	+4,4%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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CASE C04 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	60.5	60.5	
Bending moment : $M(G_1) =$ (kN.m)	189	189	
Shear force : $V(G_2) =$ (kN)	15.6	15.6	
Bending moment : $M(G_2) =$ (kN.m)	48.8	48.8	
Shear force : $V(Q_1) =$ (kN)	39.1	39.1	
Bending moment : $M(Q_1) =$ (kN.m)	122	122	
Shear force : $V(Q_c) =$ (kN)	13.3	13.3	
Bending moment : $M(Q_c) =$ (kN.m)	45.3	45.4	-0.22%
Shear force : $V(SHC) =$ (kN)	0	0	
Bending moment : $M(SHC) =$ (kN.m)	112.5	109.4	
Shear force : $V_{Ed} (ULS01) =$ (kN)	161	161	
Bending moment : $M_{Ed} (ULS01) =$ (kN.m)	504	504	
Shear force : $V_{Ed,const} (ULSC01) =$ (kN)	102	102	
Bending moment : $M_{Ed,const} (ULSC01) =$ (kN.m)	323	323	

CASE C04 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.42	0.426	
Resistance to shear force : $\Gamma_V =$	0.16	0.156	
Resistance to MV interaction : $\Gamma_{MV} =$	0.42	0.426	
Degree of shear connection : $\eta =$	0.78	0.773	
Minimum degree of connection : $\eta_{min} =$	0.63	0.625	
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2.20	2.14	2.80%
Min. compression strut inclination angle : $\theta_{f,min} =$ (°)	27	27	
Resistance of the compression strut : $\Gamma_{sf} =$	0.61	0.59	3.39%
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	2.12	2.05	3.41%

CASE C04 RESULTS: ULS AT CONSTRUCTION STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.42	0.42	
Resistance to shear force : $\Gamma_V =$	0.10	0.10	
Resistance to MV interaction : $\Gamma_{MV} =$	0.42	0.42	
Critical moment to lateral torsional buckling : M_{cr} (kN.m)	903	914	-1.20%
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	0.66	0.63	4.76%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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CASE C04 RESULTS: SLS ASSUMING FULL CONNECTION	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	31.1	31.1	
Deflection : $\delta(G_2)$ (mm)	3.79	3.79	
Deflection : $\delta(Q_1)$ (mm)	6.62	6.64	
Deflection : $\delta(Q_c)$ (mm)	6.5	6.5	
Deflection : $\delta(SHC)$ (mm)	10.3	10.1	1.98%
Deflection : $\delta(SLS)$ (mm)	51.8	51.6	0.58%
Eigen frequency : $f(G + 0.2Q_1)$ (Hz)	4.6	4.7	-2.54%

5.6 CASE C05: Composite beam with cantilever span

5.6.1 Description

This case assesses a composite beam with a cantilever span at the left end.

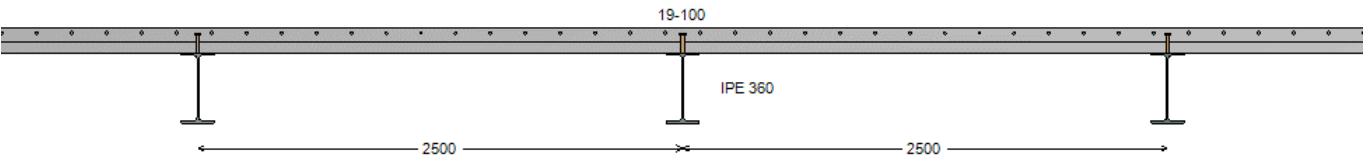


Figure 5 : Transversal view

5.6.2 Parameters

Definition of the beam:

Span lengths		
Main span L = 10.00 m		
☒ Left cantilever span L _g = 3.00 m		
☐ Right cantilever span		
☐ The slab is continuous at the right support		
Spacings		Slab openings
☒ Intermediate beam		☐ Left opening
☐ Edge beam		☐ Right opening
Spacings d ₁ = 2.50 m d ₂ = 2.50 m		

Definition of the slab:

- Composite slab
- Total depth: $t_d = 130 \text{ mm}$
- Perpendicular ribs
- Steel profiled sheeting: Cofraplus 60, thickness 0,75 mm
- Prepunched sheeting

Definition of the profile:

- Hot rolled profile
- Profile: IPE 360
- Steel grade: S355 JR

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Definition of the concrete:

Normal concrete

Concrete class: C35/45

Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 19x100

Two studs in each rib

Unpropped beam during construction

Definition of loads:

G1: self-weight only

G2: surface load 2 kN/m^2 on each span

live load Q1: surface load 16 kN/m^2 on each span

Combination of loads:

ULS: $1,35 (G1+G2) + 1,5 Q1$

SLS: $(G1+G2) + Q1$

Design parameters:

Standard: Eurocodes first generation

Design: According to the class of cross-section

Shear resistance: $\eta = 1,2$

Relative humidity: 50 %

Shrinkage deformation: 325×10^{-6}

Age of concrete: $t_0 = 28 d$ for G2

$t = 50 \times 365 d$

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5.6.3 Verification of results

CASE C05 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	72,7	72,7	
Shear area : $A_v =$ (cm ²)	35,14	35,14	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	16 270	16 266	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 043	1 041	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	903,6	903,6	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	122,8	122,5	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 019	1 019	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	191,1	188,9	
Torsional constant : $I_t =$ (cm ⁴)	37,32	37,44	
Warping constant : $I_w =$ (cm ⁶)	313 600	316 580	
Mass by length unit : $m_{lin} =$ (kg/ml)	57,1	57,1	
Nominal yield strenght : $f_y =$ (MPa)	355	355	
Tensile resistance : $N_{t,Rd} =$ (kN)	2 581,9	2582	
Shear resistance : $V_{pl,Rd} =$ (kN)	720,2	720	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	361,7	362	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	320,8	321	
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	41,83	41,8	
Shear buckling resistance : $72\varepsilon/\eta =$	48,82	48,8	
Verification of web buckling required?	No	No	

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CASE C05 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	144	144	
Char. value of the compression strength : $f_{ck} =$ (MPa)	35	35	
Mean value of compression strength : $f_{cm} =$ (MPa)	43	43	
Mean tensile strength : $f_{ctm} =$ (MPa)		3,21	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	34 080	34 077	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	44.37	44,37	
Shear resistance of the connector : $P_{Rd} =$ (kN)	41,13	41,13	
Reduction factor : $k_t =$	0,50	0,50	

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CASE C05 RESULTS: ANALYSIS FOR LOAD CASES	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(G_1) =$ (kN)	35,88	35,88	
Shear force at support B – central span : $V(G_1) =$ (kN)	29,96	29,96	
Bending moment at support A : $M(G_1) =$ (kN.m)	-29,6	-29,6	
Max. bending moment in central span : $M(G_1) =$ (kN.m)	68,14	68,14	
Shear force at support A – central span : $V(G_2) =$ (kN)	27,25	27,25	
Shear force at support B – central span : $V(G_2) =$ (kN)	22,75	22,75	
Bending moment at support A : $M(G_2) =$ (kN.m)	-22,50	-22,50	
Max. bending moment in central span : $M(G_2) =$ (kN.m)	51,75	51,75	
Shear force at support A – central span : $V(Q_1\#1) =$ (kN)	218	218	
Shear force at support B – central span : $V(Q_1\#1) =$ (kN)	182	182	
Bending moment at support A : $M(Q_1\#1) =$ (kN.m)	-180	-180	
Max. bending moment in central span : $M(Q_1\#1) =$ (kN.m)	414	414	
Shear force at support A – central span : $V(Q_1\#2) =$ (kN)	200	200	
Shear force at support B – central span : $V(Q_1\#2) =$ (kN)	200	200	
Bending moment at support A : $M(Q_1\#2) =$ (kN.m)	0	0	
Max. bending moment in central span : $M(Q_1\#2) =$ (kN.m)	500	500	
Shear force at support A – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Shear force at support B – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Bending moment at support A : $M(Q_1\#3) =$ (kN.m)	-180,0	-180,0	
Max. bending moment in central span : $M(Q_1\#3) =$ (kN.m)	0	0	

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CASE C05 RESULTS: ANALYSIS FOR ULS COMBINATION	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(ULS_1\#1) =$ (kN)	412,23	412,2	
Shear force at support B – central span : $V(ULS_1\#1) =$ (kN)	344,15	344,2	
Bending moment at support A : $M(ULS_1\#1) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#1) =$ (kN.m)	782,86	782,9	
Shear force at support A – central span : $V(ULS_1\#2) =$ (kN)	385,23	385,2	
Shear force at support B – central span : $V(ULS_1\#2) =$ (kN)	371,15	371,2	
Bending moment at support A : $M(ULS_1\#2) =$ (kN.m)	-70,37	-70,37	
Max. bending moment in central span : $M(ULS_1\#2) =$ (kN.m)	910,29	910,6	
Shear force at support A – central span : $V(ULS_1\#3) =$ (kN)	112,23	112,2	
Shear force at support B – central span : $V(ULS_1\#3) =$ (kN)	44,15	44,15	
Bending moment at support A : $M(ULS_1\#3) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#3) =$ (kN.m)	62,09	62,33	

CASE C05 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	1,575	1,575	
Resistance to shear force : $\Gamma_V =$	0,572	0,572	
Resistance to MV interaction : $\Gamma_{MV} =$	1,575	1,575	
Degree of shear connection ($M > 0$) : $\eta =$	0,431	0,428	
Minimum degree of connection ($M > 0$) : $\eta_{\min} =$	0,505	0,505	
Degree of shear connection ($M < 0$) : $\eta =$	1	1,03	
Minimum degree of connection ($M < 0$) : $\eta_{\min} =$	1	1	

CASE C05 RESULTS: TRANSVERSE REINFORCEMENTS IN THE MAIN SPAN	Reference	ABCPMX	Dev.
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2,66	2,66	
Min. compression strut inclination angle : $\theta_{f,\min} =$ (°)	36	36	
Resistance of the compression strut : $\Gamma_{sf} =$	0,464	0,46	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	3,20	3,20	

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5.7 CASE C05b: Composite beam with cantilever span

5.7.1 Description

This case assesses a composite beam with a cantilever span at the right end.

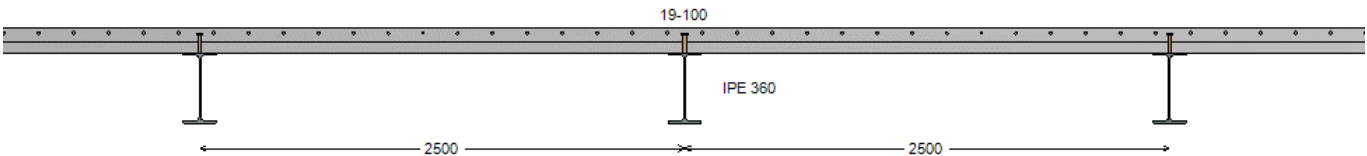


Figure 6 : Transversal view

5.7.2 Parameters

The case C05b is identical to Case C05, except that the cantilever is defined on the right end of the beam.

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5.7.3 Verification of results

CASE C05B RESULTS: ANALYSIS FOR LOAD CASES	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(G_1) =$ (kN)	29,96	29,96	
Shear force at support B – central span : $V(G_1) =$ (kN)	35,88	35,88	
Bending moment at support B : $M(G_1) =$ (kN.m)	-29,6	-29,6	
Max. bending moment in central span : $M(G_1) =$ (kN.m)	68,14	68,14	
Shear force at support A – central span : $V(G_2) =$ (kN)	22,75	22,75	
Shear force at support B – central span : $V(G_2) =$ (kN)	27,25	27,25	
Bending moment at support B : $M(G_2) =$ (kN.m)	-22,50	-22,50	
Max. bending moment in central span : $M(G_2) =$ (kN.m)	51,75	51,75	
Shear force at support A – central span : $V(Q_1\#1) =$ (kN)	182	182	
Shear force at support B – central span : $V(Q_1\#1) =$ (kN)	218	218	
Bending moment at support B : $M(Q_1\#1) =$ (kN.m)	-180	-180	
Max. bending moment in central span : $M(Q_1\#1) =$ (kN.m)	414	414	
Shear force at support A – central span : $V(Q_1\#2) =$ (kN)	200	200	
Shear force at support B – central span : $V(Q_1\#2) =$ (kN)	200	200	
Bending moment at support B : $M(Q_1\#2) =$ (kN.m)	0	0	
Max. bending moment in central span : $M(Q_1\#2) =$ (kN.m)	500	500	
Shear force at support A – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Shear force at support B – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Bending moment at support B : $M(Q_1\#3) =$ (kN.m)	-180,0	-180,0	
Max. bending moment in central span : $M(Q_1\#3) =$ (kN.m)	0	0	

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CASE C05B RESULTS: ANALYSIS FOR ULS COMBINATION	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(ULS_1\#1) =$ (kN)	344,15	344,2	
Shear force at support B – central span : $V(ULS_1\#1) =$ (kN)	412,23	412,2	
Bending moment at support B : $M(ULS_1\#1) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#1) =$ (kN.m)	782,86	782,9	
Shear force at support A – central span : $V(ULS_1\#2) =$ (kN)	371,15	371,2	
Shear force at support B – central span : $V(ULS_1\#2) =$ (kN)	385,23	385,2	
Bending moment at support B : $M(ULS_1\#2) =$ (kN.m)	-70,37	-70,37	
Max. bending moment in central span : $M(ULS_1\#2) =$ (kN.m)	910,29	910,6	
Shear force at support A – central span : $V(ULS_1\#3) =$ (kN)	44,15	44,15	
Shear force at support B – central span : $V(ULS_1\#3) =$ (kN)	112,23	112,2	
Bending moment at support B : $M(ULS_1\#3) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#3) =$ (kN.m)	62,09	62,33	

CASE C05B RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	1,575	1,575	
Resistance to shear force : $\Gamma_V =$	0,572	0,572	
Resistance to MV interaction : $\Gamma_{MV} =$	1,575	1,575	
Degree of shear connection ($M > 0$) : $\eta =$	0,431	0,428	
Minimum degree of connection ($M > 0$) : $\eta_{\min} =$	0,505	0,505	
Degree of shear connection ($M < 0$) : $\eta =$	1	1,03	
Minimum degree of connection ($M < 0$) : $\eta_{\min} =$	1	1	

CASE C05B RESULTS: TRANSVERSE REINFORCEMENTS	Reference	ABCPMX	Dev.
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2,66	2,66	
Min. compression strut inclination angle : $\theta_{f,\min} =$ (°)	36	36	
Resistance of the compression strut : $\Gamma_{sf} =$	0,464	0,46	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	3,20	3,20	

5.8 CASE C06: Composite beam with a solid slab

5.8.1 Description

This case assesses a composite beam with a solid slab.

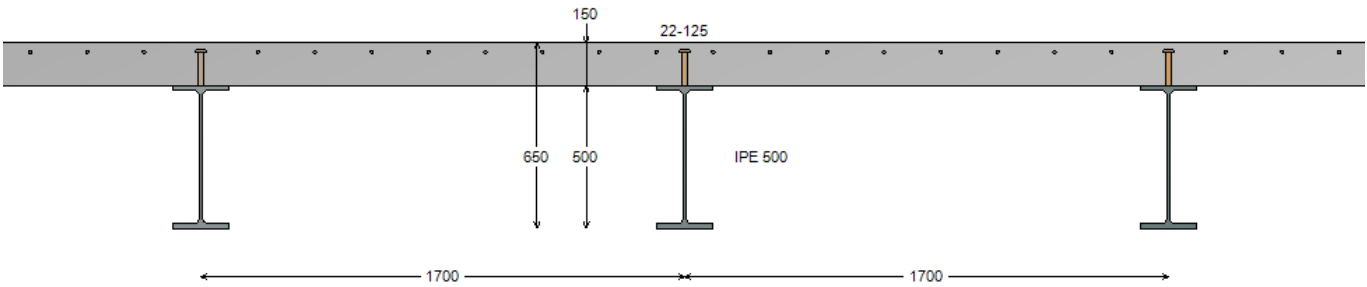


Figure 7 : Transversal view

5.8.2 Parameters

Definition of the beam:

Definition of the beam

Span lengths

Main span
L= 14,00 m

☐ Left cantilever span

☐ The slab is continuous at the left support

☐ Right cantilever span

☐ The slab is continuous at the right support

Slab openings

☐ Left opening

☐ Right opening

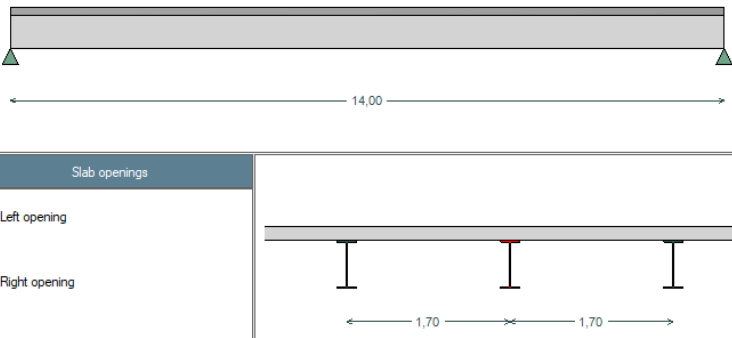
Spacings

☒ Intermediate beam

☐ Edge beam

Spacings
d₁ = 1,70 m
d₂ = 1,70 m

Diagram



Cancel

OK

Definition of the slab:

Solid slab
Total depth: $t_d = 150\text{ mm}$

Definition of the profile:

Hot rolled profile
Profile: IPE 500
Steel grade: S460 EC3

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Definition of the concrete:

Normal concrete

Concrete class: C25/30

Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 22x125

Two studs every 350 mm along the beam

The beam is fully propped during construction

Definition of loads:

G1: self-weight only

G2: surface load 2 kN/m^2

live load Q1: surface load 8 kN/m^2

Combination of loads:

ULS: $1,35 (G1+G2) + 1,5 Q1$

SLS: $(G1+G2) + Q1$

Design parameters:

Standard: Eurocodes first generation

Design: According to the class of cross-section

Relative humidity: 50 %

Shrinkage deformation: 325×10^{-6}

Age of concrete: $t_0 = 28 d$ for G2

$t = 50 \times 365 d$

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5.8.3 Verification of results

CASE C06 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	115,5	115,5	
Shear area : $A_v =$ (cm ²)	59,87	59,87	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	48 200	48 199	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 930	1 928	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	2194	2 194	
Total mass : $m =$ (kg)	1270	1270	
Nominal yield strenght : $f_y =$ (MPa)	460	460	
Tensile resistance : $N_{t,Rd} =$ (kN)	5314	5314	
Shear resistance : $V_{pl,Rd} =$ (kN)	1590	1590	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	1009	1009	
Section class for bending :	1	1	
Section class for compression :	4	4	

CASE C06 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	159	159	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 480	31 476	
Shear resistance of the connector : $P_{Rd} =$ (kN)	99,6	99,6	

CASE C06 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : $b_{eff} =$ (m)	1,7	1,7	
Modular ratio : $n(G_{1,2}) =$	26,7	26,7	
Modular ratio : $n(Q_1) =$	6,67	6,67	
Modular ratio : $n(SH) =$	25,3	25,3	
Second moment of area (y-y axis) : $I_{el,y}(n = 6.67)$ (cm ⁴)	148 643	149 064	
Second moment of area (y-y axis) : $I_{el,y}(n = 26.9)$ (cm ⁴)	105 214	105 256	
Second moment of area (y-y axis) : $I_{el,y}(n = 25.5)$ (cm ⁴)	106 945	106 974	
Plastic neutral axis : $z_{pl,y}$ (mm)	-9,4	-9,3	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	1501	1501	
Reduction factor: β	0,943	0,943	

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CASE C06 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(ULS)$ = (kN)	242,4	242,4	
Bending moment : $M(ULS)$ = (kN.m)	848,5	848,5	

CASE C06 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M =	0,565	0,565	
Resistance to shear force : Γ_V =	0,152	0,152	
Degree of shear connection : η =	1	1,1	
Potential area of shear failure a-a			
Longitudinal shear force : τ_{Ed} = (MPa)	1,84	1,84	
Min. compression strut inclination angle : $\theta_{f,min}$ = (°)	27	27	
Resistance of the compression strut : Γ_{sf} =	0,504	0,50	
Area of reinf. by length unit : A_{sf}/s_f = (cm ² /m)	3,23	3,23	
Potential area of shear failure b-b			
Longitudinal shear force : τ_{Ed} = (MPa)	1,74	1,74	
Min. compression strut inclination angle : $\theta_{f,min}$ = (°)	27	27	
Resistance of the compression strut : Γ_{sf} =	0,478	0,48	
Area of reinf. by length unit : A_{sf}/s_f = (cm ² /m)	3,23	6,67	

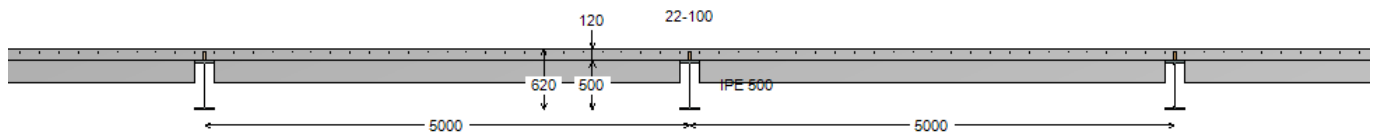
CASE C06 RESULTS: SLS ASSUMING FULL SHEAR CONNECTION	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	16,21	16,2	
Deflection : $\delta(G_2)$ (mm)	7,71	7,69	
Deflection : $\delta(Q_1)$ (mm)	21,8	21,7	

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5.9 CASE C07: Composite beam with a composite slab and Cofraplus 220

5.9.1 Description

This case assesses a composite beam with a composite slab using a Cofraplus 220.



5.9.2 Parameters

Definition of the beam:

Definition of the beam

Span lengths

Main span
L= 14.00 m

☐ Left cantilever span
☐ The slab is continuous at the left support
☐ Right cantilever span
☐ The slab is continuous at the right support

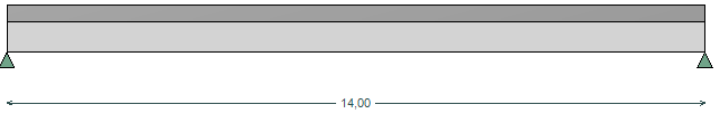
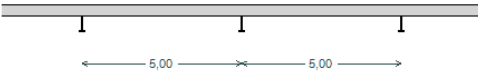
Slab openings

☐ Left opening
☐ Right opening

Spacings

☒ Intermediate beam
☐ Edge beam

Spacings
d₁ = 5.00 m
d₂ = 5.00 m

Cancel

OK

G1:	self-weight only
G2:	surface load 1 kN/m ²
live load Q1:	surface load 3 kN/m ²

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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Combination of loads:

ULS: 1,35 (G1+G2) + 1,5 Q1

SLS: (G1+G2) + Q1

Design parameters:

Standard: Eurocodes first generation

Design: According to the class of cross-section

Relative humidity: 50 %

Shrinkage deformation: 325×10^{-6}

Age of concrete: $t_0 = 28 d$ for G2

$t = 50 \times 365 d$

5.9.3 Verification of results

CASE C07 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	115,5	115,5	
Shear area : $A_v =$ (cm ²)	59,87	59,87	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	48 200	48 199	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 930	1 928	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	2194	2 194	
Total mass : $m =$ (kg)	1270	1270	
Nominal yield strenght : $f_y =$ (MPa)	420	420	
Tensile resistance : $N_{t,Rd} =$ (kN)	4 852	4 852	
Shear resistance : $V_{pl,Rd} =$ (kN)	1 452	1 452	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	921	922	
Section class for bending :	1	1	
Section class for compression :	4	4	

CASE C07 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	240	240	
Char. value of the compression strength : $f_{ck} =$ (MPa)	20	20	
Mean value of compression strength : $f_{cm} =$ (MPa)	28	28	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	29 960	29 962	
Shear resistance of the connector : $P_{Rd} =$ (kN)	86,9	86,9	
Mass of the slab: (kg/m)	1 880	1 880	
Weight of the slab: (kN/m ²)	19,26	19,26	

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CASE C07 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : b_{eff} = (m)	3,5		
Modular ratio : $n(G_{1,2})$ =	28,4	28,4	
Modular ratio : $n(Q_1)$ =	7,01	7,01	
Modular ratio : $n(SH)$ =	26,9	26,9	
Second moment of area (y-y axis) : $I_{\text{el},y}(n = 7,01)$ (cm ⁴)	148 444	148 446	
Second moment of area (y-y axis) : $I_{\text{el},y}(n = 26,9)$ (cm ⁴)	113 880	113 889	
Second moment of area (y-y axis) : $I_{\text{el},y}(n = 28,4)$ (cm ⁴)	112 303	112 330	
Plastic neutral axis : $z_{\text{pl},y}$ (mm)	0,6	0,6	
Plastic resistance to bending moment : $M_{\text{pl,Rd}}$ (kN.m)	1459	1 459	
Reduction factor: β	0,973	0,973	

CASE C07 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(ULS)$ = (kN)	395	395	
Bending moment : $M(ULS)$ = (kN.m)	1383	1383	

CASE C07 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment: Γ_{M} =	0,948	0,948	
Resistance to shear force: Γ_{V} =	0,272	0,272	
Degree of shear connection: η =	1,02	1,023	
Potential area of shear failure a-a			
Longitudinal shear force : τ_{Ed} = (MPa)	2,852	2,85	
Min. compression strut inclination angle : $\theta_{\text{f,min}}$ = (°)	27	27	
Resistance of the compression strut : Γ_{sf} =	0,958	0,96	
Area of reinf. by length unit : $A_{\text{sf}}/s_{\text{f}}$ = (cm ² /m)	4,01	4,01	
Potential area of shear failure b-b			
Longitudinal shear force : τ_{Ed} = (MPa)	2,510	2,51	
Min. compression strut inclination angle : $\theta_{\text{f,min}}$ = (°)	27	27	
Resistance of the compression strut : Γ_{sf} =	0,843	0,84	
Area of reinf. by length unit : $A_{\text{sf}}/s_{\text{f}}$ = (cm ² /m)	8,15	8,15	

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CASE C07 RESULTS: SLS ASSUMING FULL SHEAR CONNECTION	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	42,7	42,8	
Deflection : $\delta(G_2)$ (mm)	10,6	10,6	
Deflection : $\delta(Q_1)$ (mm)	24,1	24,1	

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6 TESTS CASES – STEEL BEAMS

6.1 CASE S01: Steel beam with 2 cantilever spans

6.1.1 Description

This test deals with a steel beam with both left and right cantilevers.

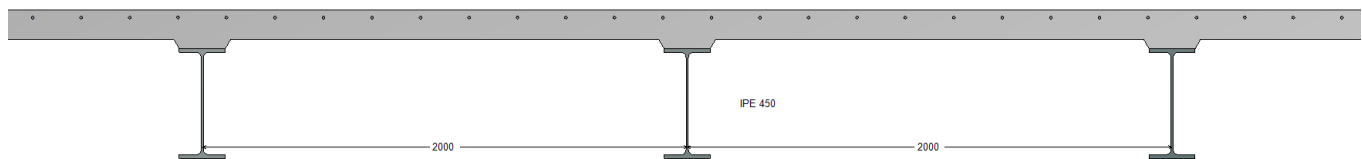
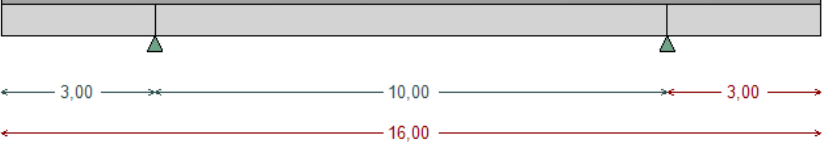
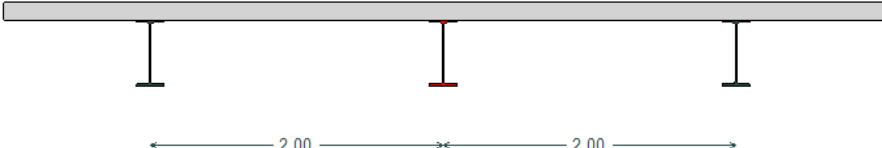


Figure 8 : Transversal view

6.1.2 Parameters

Definition of the beam:

Span lengths Main span L = 10,00 m ☒ Left cantilever span L_g = 3,00 m ☒ Right cantilever span L_d = 3,00 m	
Spacings ☒ Intermediate beam ☐ Edge beam Spacings d₁ = 2,00 m d₂ = 2,00 m	

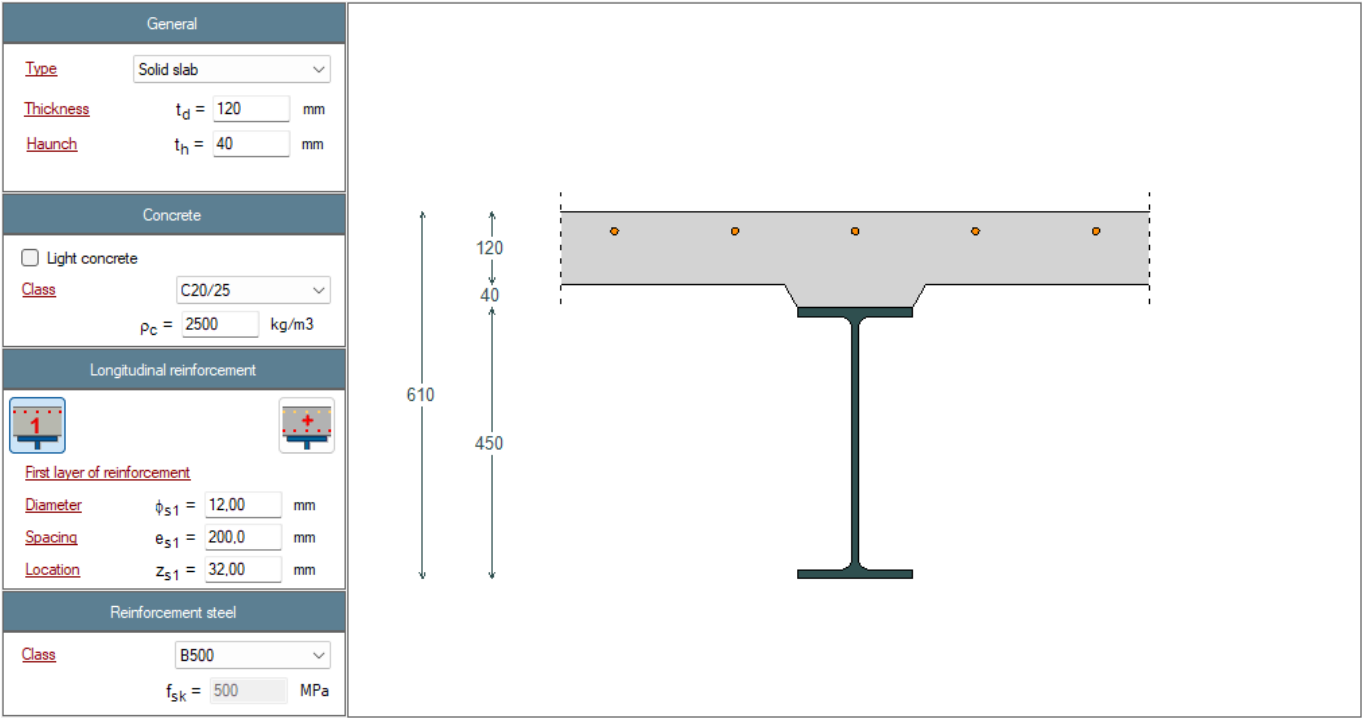
Definition of the profile:

Hot rolled profile

Profile: IPE 450

Steel grade: S355 M/ML

Definition of the slab:



Lateral restraint of the profile:

One lateral restraints at mid-span of the central span, for both flanges

Definition of loads:

G1:	self-weight only	
live load Q1:	left cantilever span:	distributed load from 0 (left end) to 1 kN/m (right end)
	central span:	distributed load from 1 (left end) to 3 kN/m (right end)
	right cantilever span:	distributed load from 3 (left end) to 5 kN/m (right end)

Definition of loads:

$\eta = 1,2$

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6.1.3 Verification of results

CASE S01 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	98.8	98.8	
Shear area : $A_v =$ (cm ²)	50.8	50.8	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	33 740	33 743	0.01%
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 676	1 672	0.24%
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 500	1 500	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	176	176	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 702	1 702	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	276	272.8	1.17%
Torsional constant : $I_t =$ (cm ⁴)	66.66	66.75	0.13%
Warping constant : $I_w =$ (cm ⁶)	780 739	791 005	1.30%
Mass by length unit : $m_{lin} =$ (kg/ml)	77.7	77.6	0.13%
Mass : $m =$ (kg)	1 243	1 241	0.16%
Total paint surface : $S =$ (m ²)	25,68	25,68	
Paint surface (without upper face) : $S =$ (m ²)	22,64	22,64	
Section factor : $M =$ (m ⁻¹)	162.4	162.4	
Section factor (without upper face) : $M =$ (m ⁻¹)	143.2	143.2	
Nominal yield strength : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	3 508	3 508	
Shear resistance : $V_{pl,Rd} =$ (kN)	1 402	1 402	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	604	604	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	532	532	
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	44,8	44,8	
Shear buckling resistance : $72\varepsilon/\eta =$	48.8	48.8	
Verification of web buckling required?	No	No	

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CASE S01 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	34.4	34.2	0.58%
Bending moment : $M(G_1) =$ (kN.m)	55.1/-31	54.8/-30.8	0.73%/0.65%
Shear force : $V(Q_1\#1) =$ (kN)	13.5	13.5	
Bending moment : $M(Q_1\#1) =$ (kN.m)	14.7/-19.5	14.7/-19.5	
Shear force : $V(Q_1\#2) =$ (kN)	11.7	11.7	
Bending moment : $M(Q_1\#2) =$ (kN.m)	25.2	25.2	
Shear force : $V(Q_1\#3) =$ (kN)	12	12	
Bending moment : $M(Q_1\#3) =$ (kN.m)	-19.5	-19.5	
Shear force : $V_{Ed,ULS\#1} =$ (kN)	66.7	66.4	0.45%
Bending moment : $M_{Ed,ULS\#1} =$ (kN.m)	96.2/-71.1	95.8/-70.8	0.52%/0.42%
Shear force : $V_{Ed,ULS\#2} =$ (kN)	64.0	63.7	0.47%
Bending moment : $M_{Ed,ULS\#2} =$ (kN.m)	112/-41.8	111/-41.6	0.90%/0.48%
Shear force : $V_{Ed,ULS\#3} =$ (kN)	49.2	48.9	0.61%
Bending moment : $M_{Ed,ULS\#3} =$ (kN.m)	59.0/-71.1	58.6/-70.8	0.85%/0.42%

CASE S01 RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS\#1} =$	0.16	0.159	
Resistance to shear force : $\Gamma_{V,ULS\#1} =$	0.06	0.064	
Resistance to MV interaction : $\Gamma_{MV,ULS\#1} =$	0.16	0.159	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS\#1}$ (kN.m)	521.97	523.4	-0.27%
Resistance to lateral torsional buckling : $\Gamma_{LT,ULS\#1} =$	0.29	0.288	
Resistance to bending moment : $\Gamma_{M,ULS\#2} =$	0.19	0.185	
Resistance to shear force : $\Gamma_{V,ULS\#2} =$	0.06	0.061	
Resistance to MV interaction : $\Gamma_{MV,ULS\#2} =$	0.19	0.185	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS\#2}$ (kN.m)	546.2	545.3	0.53%
Resistance to lateral torsional buckling : $\Gamma_{LT,ULS\#2} =$	0.33	0.327	
Resistance to bending moment : $\Gamma_{M,ULS\#3} =$	0.12	0.117	
Resistance to shear force : $\Gamma_{V,ULS\#3} =$	0.05	0.047	
Resistance to MV interaction : $\Gamma_{MV,ULS\#3} =$	0.12	0.117	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS\#3}$ (kN.m)	496.6	497.1	-0.10%
Resistance to lateral torsional buckling : $\Gamma_{LT,ULS\#3} =$	0.22	0.220	

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CASE S01 RESULTS: SLS		Reference	ABCPMX	Dev.
Deflection central span : $\delta(G_1)$	(mm)	7.19	7.14	
Deflection central span : $\delta(Q_1\#1)$	(mm)	1.83	1.83	0.00%
Deflection central span : $\delta(Q_1\#2)$	(mm)	3.68	3.68	0.00%
Deflection right span : $\delta(Q_1\#3)$	(mm)	3.50	3.50	
Deflection central span : $\delta(SLS\#1)$	(mm)	9.01	8.96	
Deflection central span : $\delta(SLS\#2)$	(mm)	10.9	10.8	
Deflection central span : $\delta(SLS\#3)$	(mm)	5.34	5.30	

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6.2 CASE S02: Steel beam with welded profile

6.2.1 Description

This test deals with a steel beam with welded profile (double symmetric).

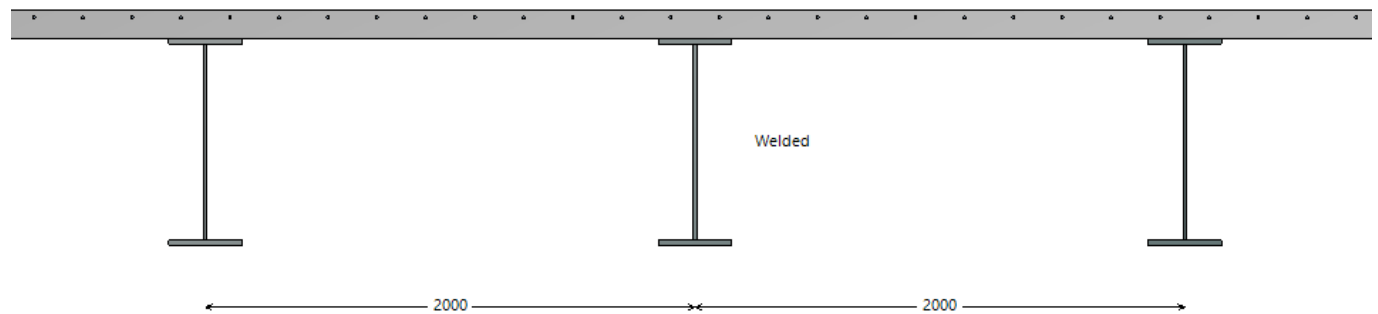


Figure 9 : Transversal view

6.2.2 Parameters

Definition of the beam:

Single span
Span length: 15 m

Definition of the profile:

Welded profile
Upper flange: 300x20
Web: 800x12
Lower flange: 300x20
Steel grade: S355 JR

Definition of loads:

G1: distributed load: 12 kN/m
live load Q1: distributed load: 20 kN/m
live load Q2: concentrated load: 15 kN at mid span

Lateral restraints:

2 lateral restraints of both flange at 1/3 and 2/3 of the span length

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Partial factors and national factors:

$$\gamma_G = 1,4$$

$$\gamma_Q = 1,6$$

$$\gamma_{M0} = 1,05$$

$$\gamma_{M1} = 1,1$$

$$\gamma_{M2} = 1,25$$

$$\psi_0 = 0,7$$

$$\eta = 1,2$$

Combination of loads:

$$\text{ULS-01:} \quad \gamma_G G + \gamma_Q Q_1 + \gamma_Q \psi_0 Q_2$$

$$\text{ULS-02:} \quad \gamma_G G + \gamma_Q \psi_0 Q_1 + \gamma_Q Q_2$$

$$\text{SLS:} \quad G + Q_1 + Q_2$$

Design parameters:

Standard: Eurocodes first generation

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6.2.3 Verification of results

CASE S02 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	216	216	
Shear area : $A_v =$ (cm ²)	115,2	115,2	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	252 960	252 960	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	9012	9012	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	6023	6023	
Mass by length unit : $m_{lin} =$ (kg/ml)	170	170	
Mass : $m =$ (kg)	2543	2543	
Total paint surface : $S =$ (cm ²)	42,84	42,84	
Paint surface (without upper face) : $S =$ (cm ²)	38,34	38,34	
Nominal yield strength flanges : $f_y =$ (MPa)	345	345	
Nominal yield strength web : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	7189	7189	
Shear resistance : $V_{pl,Rd} =$ (kN)	2249	2249	
Plastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	2266	2266	
Elastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	1979	1979	
Section class for bending :	2	2	
Section class for compression :		4	
Shear buckling resistance : $h_w/t_w =$	66,67	66,67	
Shear buckling resistance : $72\varepsilon/\eta =$	48,82	48,82	
Verification of web buckling required?	Yes	Yes	

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CASE S02 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	90	90	
Bending moment : $M(G_1) =$ (kN.m)	337,5	337,5	
Shear force : $V(Q_1) =$ (kN)	150	150	
Bending moment : $M(Q_1) =$ (kN.m)	562,5	562,5	
Shear force : $V(Q_2) =$ (kN)	7,5	7,5	
Bending moment : $M(Q_2) =$ (kN.m)	56,3	56,3	
Shear force : $V_{Ed,ULS-1} =$ (kN)	374,4	374,4	
Bending moment : $M_{Ed,ULS-1} =$ (kN.m)	1435,5	1435,5	
Shear force : $V_{Ed,ULS-2} =$ (kN)	306	306	
Bending moment : $M_{Ed,ULS-2} =$ (kN.m)	1192,5	1192,5	

CASE S02 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,634	0,634	
Resistance to shear force : Γ_V	0,166	0,166	
Resistance to shear buckling : Γ_{Vb}	0,239	0,239	
Resistance to MV interaction : Γ_{MV}	-	-	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	4163	4163	
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	1,09	1,09	
Minimum design size of welds: a_w (mm)	0,75	0,75	

CASE S02 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G)$ (mm)	14,9	14,9	
Deflection : $\delta(Q_1)$ (mm)	24,8	24,8	
Deflection : $\delta(Q_2)$ (mm)	1,99	1,98	
Deflection : $\delta(SLS)$ (mm)	41,7	41,7	
Eigen frequency : $f(G + 0,2 Q_1)$ (Hz)	3,98	3,98	
Modal mass: (kg)	12 232	12 232	

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6.3 CASE S02b: Steel beam with welded profile

6.3.1 Description

This test deals with a steel beam with welded profile (double symmetric). An elastic design is triggered by the class 3 section

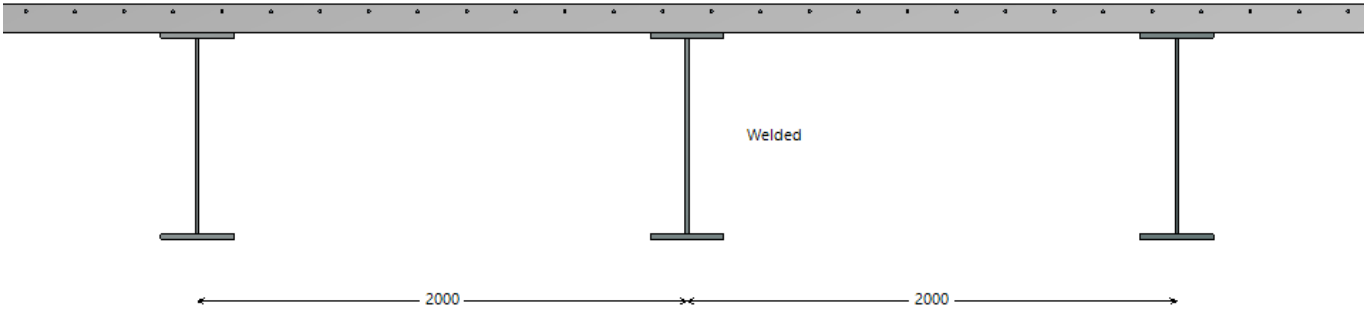


Figure 10 : Transversal view

6.3.2 Parameters

All parameters are the same as in Case S02, with the following modifications:

Definition of the profile:

Welded profile	
Upper flange:	300x20
Web:	800x12
Lower flange:	300x20
Steel grade:	S460 JR

Definition of loads:

G1:	distributed load:	12 kN/m
live load Q1:	distributed load:	20 kN/m
live load Q2:	concentrated load:	30 kN at mid span

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
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6.3.3 Verification of results

CASE S02B RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	216	216	
Shear area : $A_v =$ (cm ²)	115,2	115,2	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	252 960	252 960	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	9012	9012	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	6023	6023	
Mass by length unit : $m_{lin} =$ (kg/ml)	170	170	
Mass : $m =$ (kg)	2543	2543	
Total paint surface : $S =$ (cm ²)	42,84	42,84	
Paint surface (without upper face) : $S =$ (cm ²)	38,34	38,34	
Nominal yield strength flanges : $f_y =$ (MPa)	440	440	
Nominal yield strength web : $f_y =$ (MPa)	460	460	
Axial resistance : $N_{t,Rd} =$ (kN)	9234	9234	
Shear resistance : $V_{pl,Rd} =$ (kN)	2914	2914	
Plastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	2524	2524	
Elastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	2903	2903	
Section class for bending :	3	3	
Shear buckling resistance : $h_w/t_w =$	66,67	66,67	
Shear buckling resistance : $72\varepsilon/\eta =$	42,89	42,89	
Verification of web buckling required?	Yes	Yes	

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CASE S02B RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	90	90	
Bending moment : $M(G_1) =$ (kN.m)	337,5	337,5	
Shear force : $V(Q_1) =$ (kN)	150	150	
Bending moment : $M(Q_1) =$ (kN.m)	562,5	562,5	
Shear force : $V(Q_2) =$ (kN)	15	15	
Bending moment : $M(Q_2) =$ (kN.m)	112,5	112,5	
Shear force : $V_{Ed,ULS-1} =$ (kN)	382,8	382,8	
Bending moment : $M_{Ed,ULS-1} =$ (kN.m)	1498,5	1498,5	
Shear force : $V_{Ed,ULS-2} =$ (kN)	318	318	
Bending moment : $M_{Ed,ULS-2} =$ (kN.m)	1282,5	1282,5	

CASE S01 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,594	0,594	
Resistance to shear force : Γ_V	0,131	0,131	
Resistance to shear buckling : Γ_{Vb}	0,215	0,215	
Resistance to MV interaction : Γ_{MV}	-	-	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	4189	4189	
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	1,07	1,07	
Minimum design size of welds: a_w (mm)	0,75	0,73	

CASE S02 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G)$ (mm)	14,9	14,9	
Deflection : $\delta(Q_1)$ (mm)	24,8	24,8	
Deflection : $\delta(Q_2)$ (mm)	3,97	3,97	
Deflection : $\delta(SLS)$ (mm)	43,7	43,7	
Eigen frequency : $f(G + 0,2 Q_1)$ (Hz)	3,98	3,98	
Modal mass: (kg)	12 232	12 232	

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	Product reference: DRV/PMX/MV/001						
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6.4 CASE S03: Steel beam with welded profile and MV interaction

6.4.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

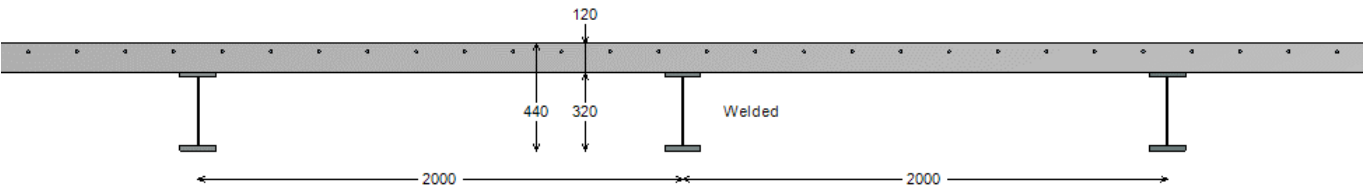


Figure 11 : Transversal view

6.4.2 Parameters

Definition of the beam:

- Single span with a cantilever span on the left
- Span length: 9 m
- Cantilever length: 2,5 m

Definition of the profile:

- Welded profile
- Upper flange: 150x18
- Web: 280x6
- Lower flange: 150x22
- Steel grade: S275 JR

Definition of the slab:

- solid slab: $t_d = 120\text{ mm}$

Definition of loads:

- live load Q1: distributed load on the cantilever span: 75 kN/m

Lateral restraints:

- 2 lateral restraints of both flange at 1/3 and 2/3 of the span length

Partial factors and national factors:

- $\gamma_{M0} = 1,0$
- $\gamma_{M1} = 1,0$
- $\gamma_{M2} = 1,25$

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$$\eta = 1,0$$

Combination of loads:

ULS-01: Q_1

SLS: Q_1

Design parameters:

Standard: Eurocodes first generation

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6.4.3 Verification of results

CASE S03 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	76,80	76,80	
Shear area : $A_v =$ (cm ²)	16,8	16,8	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	14 517	14 517	
Location of the elastic neutral axis: $z_{y,ENA} =$ (mm)	170,5	170,5	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 125	1 126	
Torsion constant : $I_t =$ (cm ⁴)	78,91	78,91	
Warping factor : $I_w =$ (cm ⁶)	250593	250594	
Mass by length unit : $m_{lin} =$ (kg/ml)	60	60	
Mass : $m =$ (kg)	693	693	
Total paint surface : $S =$ (cm ²)	14,12	14,12	
Paint surface (without upper face) : $S =$ (cm ²)	12,40	12,40	
Nominal yield strength flanges : $f_y =$ (MPa)	275	275	
Nominal yield strength web : $f_y =$ (MPa)	275	275	
Axial resistance : $N_{t,Rd} =$ (kN)	2 112	2 112	
Shear resistance : $V_{pl,Rd} =$ (kN)	267,0	267	
Plastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	276	276	
Location of the plastic neutral axis: $z_{y,PNA} =$ (mm)	208	208	
Elastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	234	234	
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	46,67	46,7	
Shear buckling resistance : $72\varepsilon/\eta =$	66,56	66,56	
Verification of web buckling required?	No	No	

CASE S03 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1) =$ (kN)	187,5	187,5	
Bending moment : $M(Q_1) =$ (kN.m)	-234,4	-234,4	

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CASE S03 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,850	0,850	
Resistance to shear force : Γ_V	0,703	0,703	
Resistance to shear buckling : Γ_{Vb}	-	-	
Resistance to MV interaction : Γ_{MV}	0,869	0,869	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	632,5	642,8	+1,6%
Resistance to lateral torsional buckling : Γ_{LT} =	1,27	1,26	
Minimum design size of welds upper flange: a_w (mm)	1,40	1,40	
Minimum design size of welds lower flange: a_w (mm)	1,47	1,47	

CASE S03 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever : $\delta(Q_1)$ (mm)	69,7	69,7	
Deflection at mid span : $\delta(Q_1)$ (mm)	-39,9	-39,9	

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6.5 CASE S03b: Steel beam with welded profile and MV interaction

6.5.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

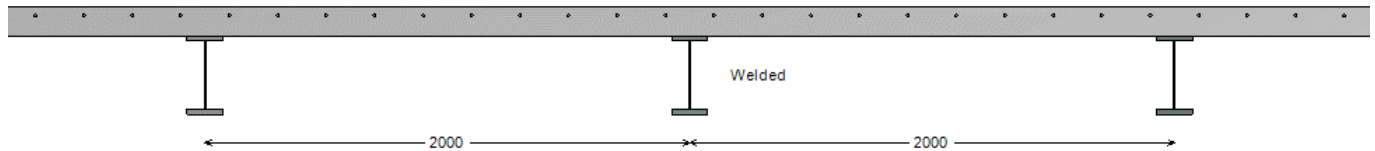


Figure 12 : Transversal view

6.5.2 Parameters

The case S03b is identical to Case S03, except that the cantilever is defined on the right end of the beam.

6.5.3 Verification of results

CASE S03B RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1) =$ (kN)	187,5	187,5	
Bending moment : $M(Q_1) =$ (kN.m)	-234,4	-234,4	

CASE S03B RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,850	0,850	
Resistance to shear force : Γ_V	0,703	0,703	
Resistance to shear buckling : Γ_{Vb}	-	-	
Resistance to MV interaction : Γ_{MV}	0,869	0,869	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	632,5	647,8	+2,4%
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	1,27	1,26	
Minimum design size of welds upper flange: a_w (mm)	1,40	1,40	
Minimum design size of welds lower flange: a_w (mm)	1,47	1,47	

CASE S03B RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever : $\delta(Q_1)$ (mm)	69,7	69,7	
Deflection at mid span : $\delta(Q_1)$ (mm)	-39,9	-39,9	

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6.6 CASE S04: Steel beam with welded profile and interaction between bending and shear buckling

6.6.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

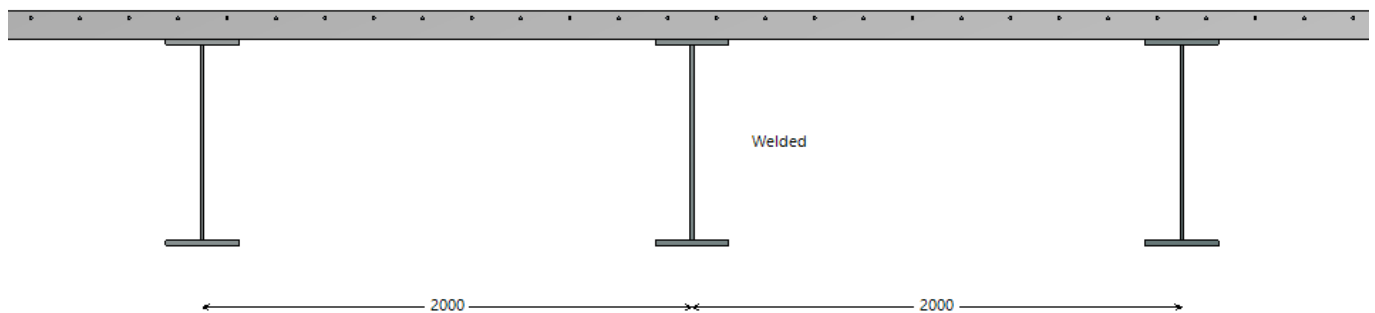


Figure 13 : Transversal view

6.6.2 Parameters

Definition of the beam:

Single span with a cantilever span on the left

Span length: 9 m

Cantilever length: 2,5 m

Definition of the profile:

Welded profile

Upper flange: 150x18

Web: 280x5

Lower flange: 150x22

Steel grade: S460 EC3

Definition of loads:

live load Q_l: distributed load on the cantilever span: 120 kN/m

Lateral restraints:

2 lateral restraints of both flange at 1/3 and 2/3 of the span length

Partial factors and national factors:

$$\gamma_{M0} = 1,0$$

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$$\gamma_{M1} = 1,0$$

$$\gamma_{M2} = 1,25$$

$$\eta = 1,2$$

Combination of loads:

$$\text{ULS-01:} \quad Q_1$$

$$\text{SLS:} \quad Q_1$$

Design parameters:

Standard: Eurocodes first generation

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6.6.3 Verification of results

CASE S04 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	74	74	
Shear area : $A_v =$ (cm ²)	16,8	16,8	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	14 329	14 329	
Location of the elastic neutral axis: $z_{y,ENA} =$ (mm)	171,0	171	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 125,3	1 125	
Mass by length unit : $m_{lin} =$ (kg/ml)	58,1	58,1	
Mass : $m =$ (kg)	668	668	
Total paint surface : $S =$ (cm ²)	14,15	14,15	
Paint surface (without upper face) : $S =$ (cm ²)	12,42	12,42	
Nominal yield strength flanges : $f_y =$ (MPa)	460	460	
Nominal yield strength web : $f_y =$ (MPa)	460	460	
Axial resistance : $N_{t,Rd} =$ (kN)	3 404	3 404	
Shear resistance : $V_{pl,Rd} =$ (kN)	446,2	446,2	
Plastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	451	451	
Location of the plastic neutral axis: $z_{y,PNA} =$ (mm)	218	218	
Elastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	386	386	
Section class for bending :	3		
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	56,0	56,0	
Shear buckling resistance : $72\varepsilon/\eta =$	42,9	42,9	
Verification of web buckling required?	Yes	Yes	

CASE S04 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1) =$ (kN)	300	300	
Bending moment : $M(Q_1) =$ (kN.m)	-375	-375	

	Project: ABCPMX-II Software - DAL-22-003						
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CASE S04 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,973	0,973	
Resistance to shear force : Γ_V	0,672	0,672	
Resistance to shear buckling : Γ_{Vb}	0,881	0,881	
Resistance to MV interaction : Γ_{MV}	0,933	0,933	

CASE S04 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever : $\delta(Q_1)$ (mm)	112,9	112,9	
Deflection at mid span : $\delta(Q_1)$ (mm)	-64,8	-64,8	

	Project: ABCPMX-II Software - DAL-22-003						
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6.7 CASE S04b: Steel beam with welded profile and interaction between bending and shear buckling

6.7.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

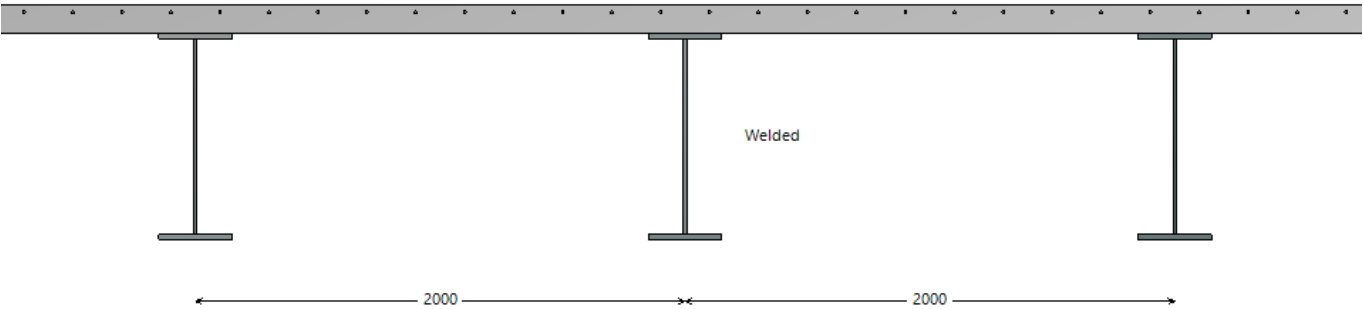


Figure 14 : Transversal view

6.7.2 Parameters

The case S04b is identical to Case S04, except that the cantilever is defined on the right end of the beam.

6.7.3 Verification of results

CASE S04B RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1)$ = (kN)	300	300	
Bending moment : $M(Q_1)$ = (kN.m)	-375	-375	

	Project: ABCPMX-II Software - DAL-22-003						
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CASE S04 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,973	0,973	
Resistance to shear force : Γ_V	0,672	0,672	
Resistance to shear buckling : Γ_{Vb}	0,881	0,881	
Resistance to MV interaction : Γ_{MV}	0,933	0,933	

CASE S04 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever: $\delta(Q_1)$ (mm)	112,9	112,9	
Deflection at mid span: $\delta(Q_1)$ (mm)	-64,8	-64,8	